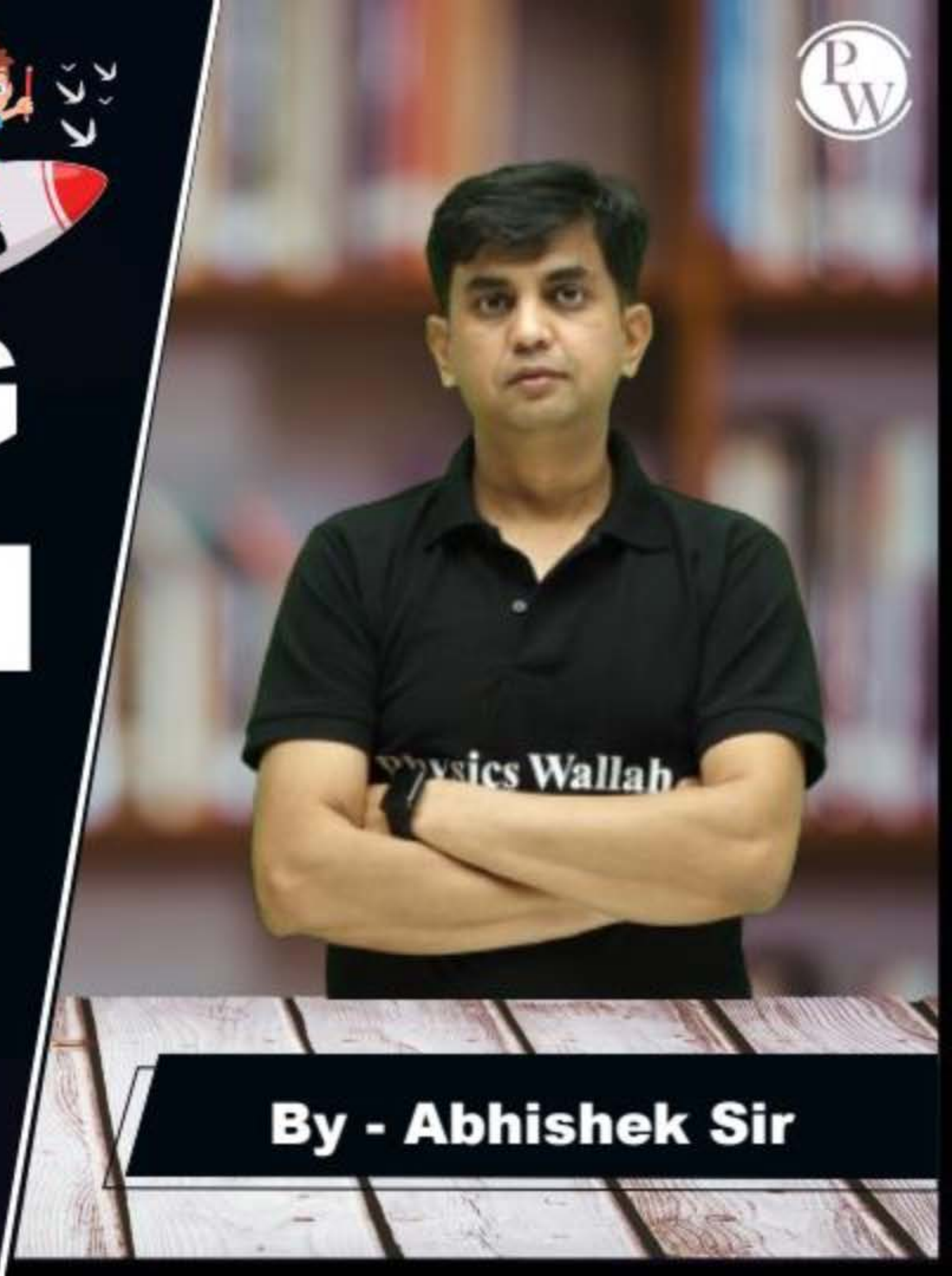
CS & IT ENGINEERING

Computer Network

Transport Layer



Lecture No. - 02

By - Abhishek Sir

Recap of Previous Lecture

Topic UDP

Topics to be Covered

Topic UDP

Topic TCP

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : UDP

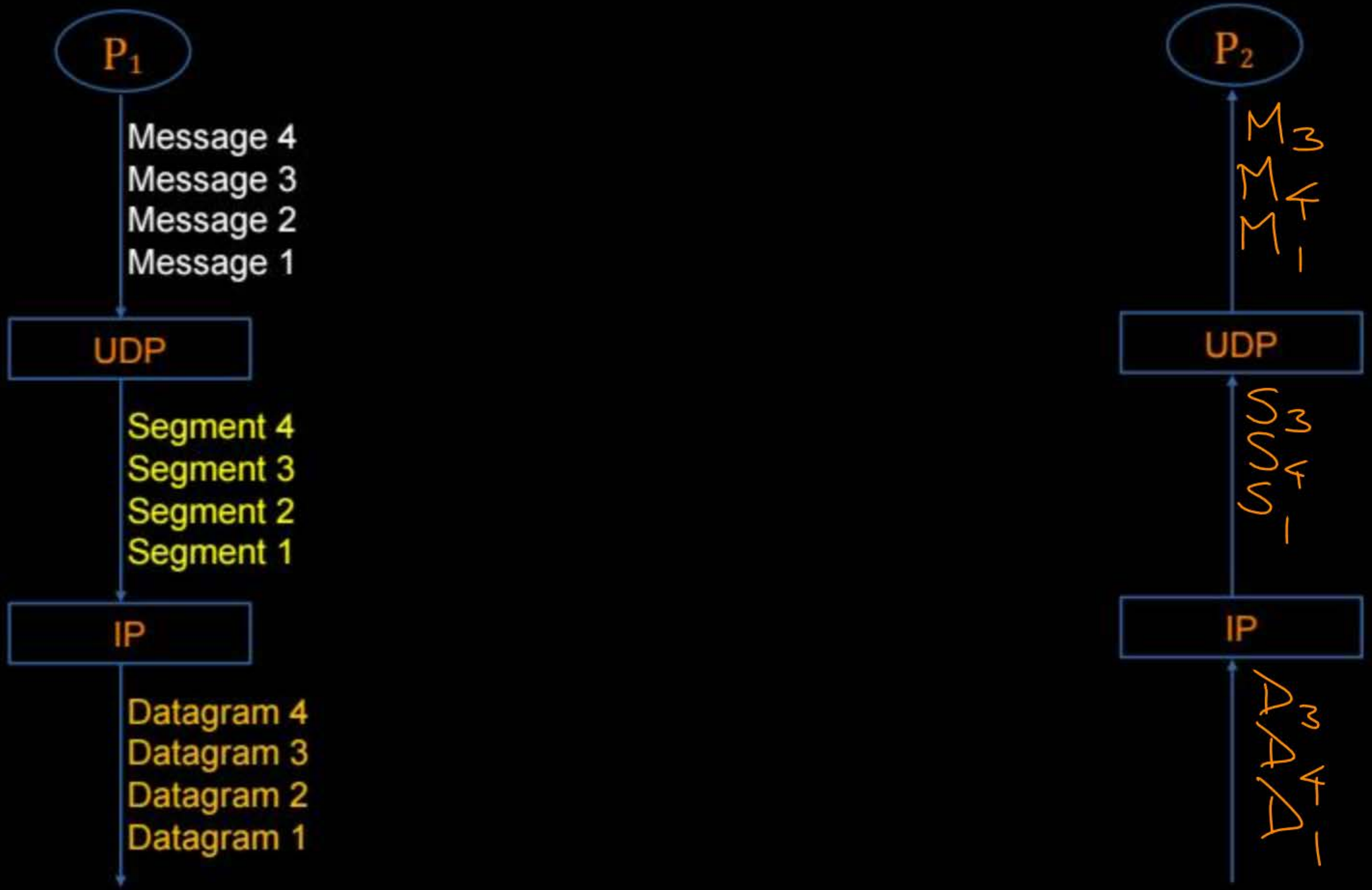
–> UDP : User Datagram Protocol

–> Provide 'Connection-less' and 'Unrealiable' services
[Unordered delivery of messages, messages may be lost]

IP also provide same services.

–> Prefer for shorter communication [like query and response]

–> UDP as transport protocol used by :
DNS, SNMP, HTTP/3, RIP and Real Time Multimedia [Streaming Multimedia]



P1
Message 4
Message 3
Message 2
Message 1
UDP
Segment 4
Segment 3
Segment 2
Segment 1
IP
Datagram 4
Datagram 3
Datagram 2
Datagram 1
P2
M3
M4
M1
UDP
S3
S4
S1
IP
D3
D4
D1

Topic : UDP

=> Connection-less :

-> No handshaking between UDP sender and receiver

-> Each UDP segment handled independently of others → UDP is stateless protocol.

-> Delivery of messages can be any order to the communicating process

=> Unrealiable :

-> Messages may be lost [ACK does not exist]

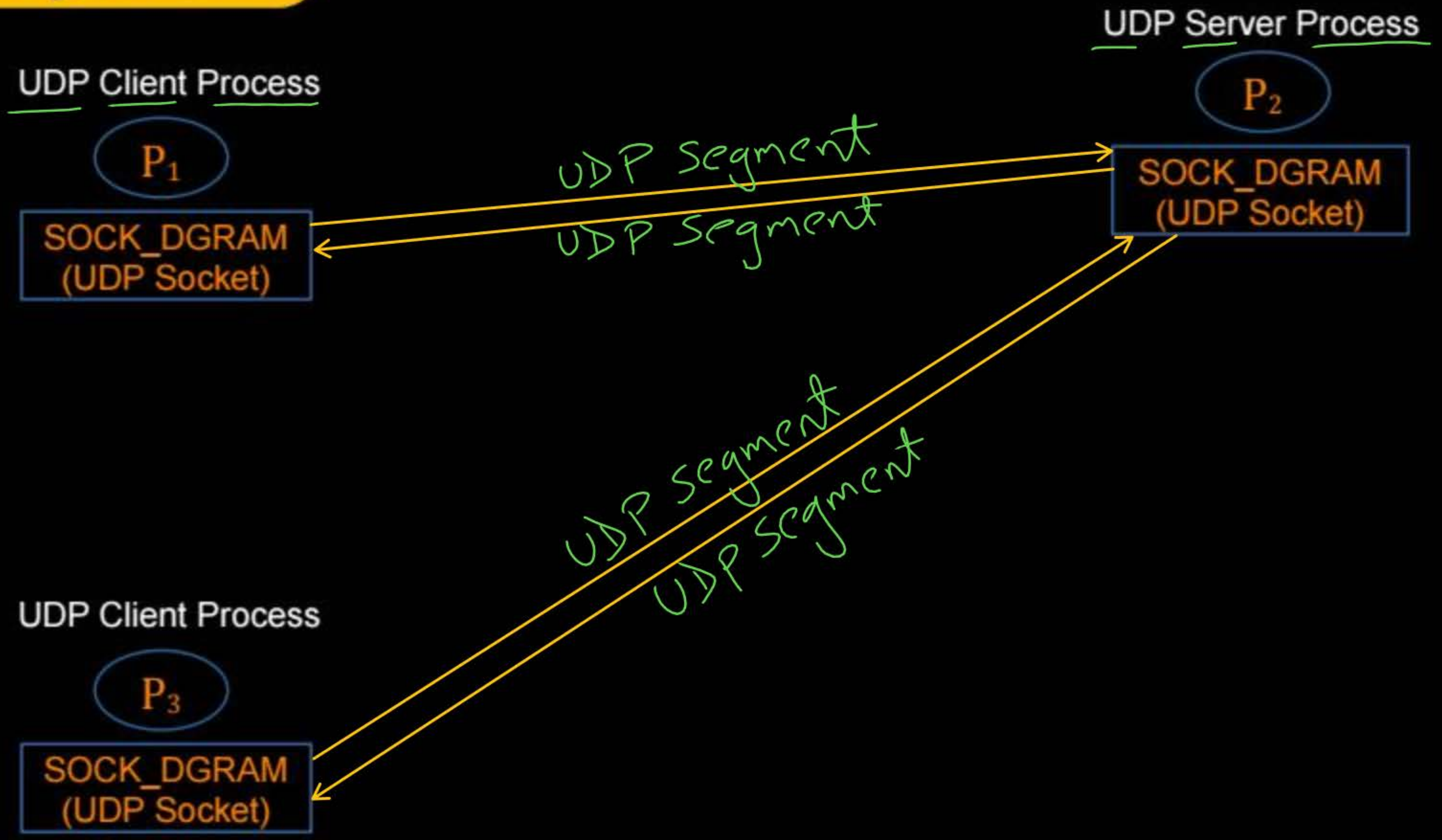
=> Simple and Fast :

-> No connection state at sender and receiver ⟹ stateless protocol

-> Small (8 bytes, fixed) header size

-> No congestion control

Topic : UDP


UDP Server Process
P_2
SOCK_DGRAM
(UDP Socket)
UDP Client Process
P_1
SOCK_DGRAM
(UDP Socket)
UDP segment
UDP segment
UDP segment
UDP segment
UDP Client Process
P_3
SOCK_DGRAM
(UDP Socket)

Topic : UDP Header

0 — 15	16 — 31
Source Port No. (16 bit)	Dest port No. (16 bit)
Total UDP length (16 bit)	Checksum (16 bit)
— // — Payload — // —	

UDP Header (8 byte)

UDP Segment

Topic : UDP Header

=> Source port number : 16-bits

=> Destination port number : 16-bits

=> Total Length : 16-bits

-> Size of UDP segment in bytes

-> Including 8 bytes header size

-> Maximum UDP segment size = $(2^{16} - 1)$ bytes

-> Maximum UDP payload size = $(2^{16} - 9)$ bytes

Topic : IPv4 Pseudo-Header

0	16	31
Source IP Address (32 bit)		
Dest. IP Address (32 bit)		
Reserved	Protocol Type	Total UDP length (16 bit)

12 byte

Topic : IPv4 Pseudo-Header

=> Source IP Address : 32-bits

=> Destination IP Address : 32-bits

=> Reserved : 8-bits

-> Must be zero

=> Protocol : 8-bits

-> Same as IP protocol type

=> UDP Length : 16-bits

-> Size of UDP segment in bytes

-> 8 bytes header size plus payload size

Topic : UDP Header

IPv4 Pseudo Header (12 byte)

UDP Header (8 byte)

UDP Segment

<table>
<tr><th>0</th><th></th><th>16</th><th>31</th></tr>
<tr><td colspan="4">Source IPv4 Address (32 bits)</td></tr>
<tr><td colspan="4">Destination IPv4 Address (32 bits)</td></tr>
<tr><td>Reserved</td><td>Protocol</td><td colspan="2">UDP Length (16 bits)</td></tr>
<tr><td colspan="2">Source Port Number (16 bits)</td><td colspan="2">Destination Port Number (16 bits)</td></tr>
<tr><td colspan="2">Total Length (16 bits)</td><td colspan="2">Checksum (16 bits)</td></tr>
<tr><td colspan="4">—//— payload —//—</td></tr>
</table>

Topic : UDP Header

=> Checksum : 16-bits

–> 16-bit one's complement of the one's complement sum of all 16-bit words in UDP segment including pseudo header

–> Error detection is optional in UDP, it depends upon the application

–> If application wants to disable error detection process then sender set all the 16 bits are zero of checksum field

Topic : TCP

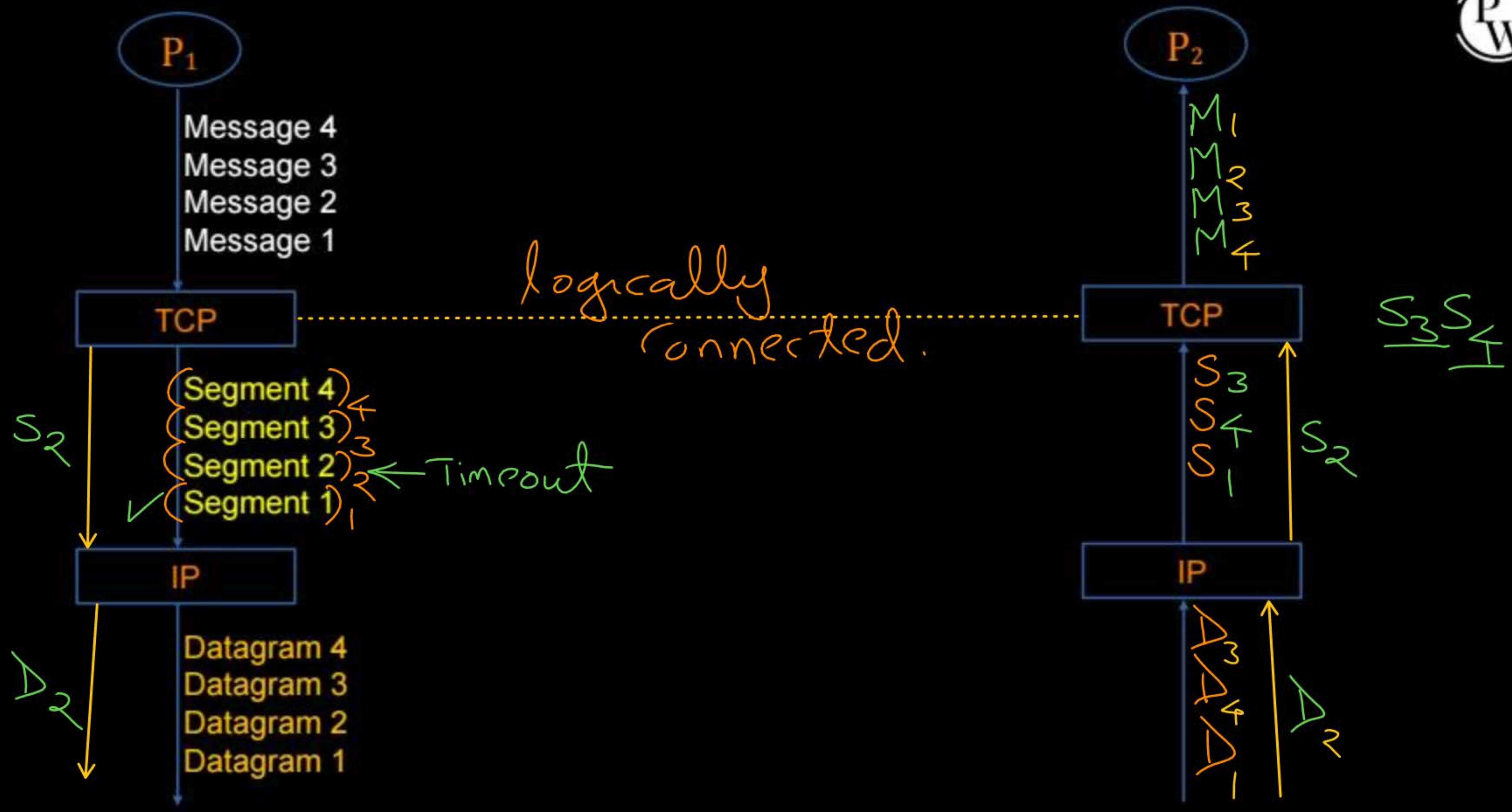
–> TCP : Transmission Control Protocol

–> Provide 'Connection-oriented' and 'Realiable' services [In-order delivery of messages]

–> Full-duplex and point-to-point logical connection

–> Provide Flow, Error and Congestion Control

–> Prefer for long communication



P_1
P_2
Message 4
Message 3
Message 2
Message 1
TCP
TCP
logically connected.
Segment 4
Segment 3
Segment 2
Segment 1
Timeout
IP
IP
Datagram 4
Datagram 3
Datagram 2
Datagram 1
S_2
D_2
M_1
M_2
M_3
M_4
S_3
S_4
S_1
S_2
$S_3 S_4$
D_3
D_4
D_1
D_2

Topic : TCP

=> Connection-oriented :

-> Need to establish (logical) connection between TCP sender and receiver

- 3-way handshaking for connection establishment
- 4-way handshaking for connection close

-> In order delivery of messages to the communicating process

=> Realiable :

-> Recover lost messages, via retransmission

=> Complex and Slow :

-> Need to maintain connection state at TCP sender and receiver

-> Provide flow and congestion control

-> TCP is stateful protocol.

2 mins Summary

Topic UDP ✓

Topic TCP

THANK - YOU

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